Advanced Machine Learning

Lab 6 --- Module 4

Instructions on how to submit your work on Canvas

To convert your Notebook to an HTML file, follow these steps:

- Open the Notebook that you want to convert to an HTML file.
- Ensure that the notebook is saved with the latest changes.
- In the Notebook interface, navigate to File in the menu bar.
- Click on Download as and select HTML (.html) from the dropdown menu. This will initiate the conversion process.
- The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
- Locate the downloaded HTML file on your computer and verify that the conversion was successful.
- Optionally, open the HTML file in a web browser to view and verify the content.
- Upload your HTML file on Canvas.

Make sure to save your Notebook before attempting to convert it to an HTML file.

Exercise 1 (5 points)

1. Load the MNIST dataset and split it into a training set and a test set (take the first 60,000 instances for training, and the remaining 10,000 for testing).
2. Train a random forest classifier on the original dataset and time how long it takes. Evaluate the resulting model on the test set.
3. Use PCA to reduce the dataset's dimensionality, with an explained variance ratio of 95%. Train a new random forest classifier on the reduced dataset and time how long it takes. Evaluate the classifier on the test set. Was training much faster? How does the performance compare to the previous classifier?
4. Train an SGDClassifier on the original dataset. Evaluate the classifier on the test set. Time how long it takes to train the classifier. Next, use the PCA-reduced dataset and train a new SGDClassifier. Evaluate and time the classifier. How much does PCA help in this case?
5. Introduce kernel PCA:

- Use a radial basis function (RBF) kernel to perform kernel PCA on the dataset. Set the gamma parameter to a value that makes the transformation effective.
    - Reduce the dataset's dimensionality to 2 dimensions using kernel PCA. Visualize the result with a scatter plot.
    - Use kernel PCA with an RBF kernel to reduce the dataset's dimensionality while preserving 95% of the variance.
6. Train a random forest classifier on the kernel PCA-reduced dataset and time how long it takes. Evaluate the resulting model on the test set. Compare the training time and performance to the previous classifiers.
7. Train an SGDClassifier on the kernel PCA-reduced dataset and time how long it takes. Evaluate the resulting model on the test set. Compare the results to the previous classifiers.

Expected Results:

- Compare the training times and performances of the classifiers (Random Forest and SGDClassifier) on the original, PCA-reduced, and kernel PCA-reduced datasets.
- Discuss the effectiveness of PCA and kernel PCA in reducing training time and improving or maintaining classifier performance.
- Visualize and interpret the results of the kernel PCA scatter plot.

Hints:

- Use `KernelPCA` from `sklearn.decomposition` for kernel PCA.
- Use `RandomForestClassifier` and `SGDClassifier` from `sklearn.ensemble` and `sklearn.linear_model` respectively.
- For timing, use the `time` module.
- For visualization, use `matplotlib` or `seaborn`.